

Shashank Kammanahalli Chandra Sekhara

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SUMMARY

Data Scientist with 2.7 years turning ambiguous, real-world problems into forecasting, classification, and anomaly-detection systems that ship. First-author on a 2026 IEEE Access paper on smart-building energy prediction (XGBoost $R^2 = 0.98$ across 91 buildings, ~\$600K/year forecasted savings), currently a Research Intern at CSU where disciplined metric slicing turned a GMM soft-clustering negative result into a simpler per-meter feature model (test R^2 0.738) that stays stable across time splits where the clustering pipeline collapses. Fluent across time-series forecasting, audio deep learning, and Transformer-internals research, with the statistical rigor (paired significance tests, bootstrap confidence intervals, controlled ablations) that keeps model claims defensible in peer review and in front of stakeholders.

SKILLS

Machine Learning & Modeling: XGBoost, LightGBM, scikit-learn, PyTorch, GMM Soft-Clustering, LSTM, Random Forest, Prophet, STL / MSTL Decomposition, tslearn, DTW, EfficientNet / timm, Transfer Learning, Multi-label Classification, Feature Engineering, Cross-Validation, Hyperparameter Tuning

Statistics & Evaluation: Time-Series Forecasting, Anomaly / Fault Detection, Clustering, Hypothesis Testing, Bootstrap Confidence Intervals, Paired Wilcoxon Tests, Ablation Studies, Controlled Experimental Design, Model Evaluation & Monitoring, Isotonic Calibration, QuantileTransformer, Fourier Features

Programming: Python, SQL, Scala, Java, Jupyter Notebooks

Data & Platforms: Pandas, NumPy, PySpark, Apache Spark, PostgreSQL, Redis, Kafka, GeoPandas, MLflow, Azure Databricks, AWS (S3, EMR, Lambda), Docker, Airflow, Git

Visualization & Apps: Matplotlib, Seaborn, Plotly, Power BI, Streamlit, FastAPI

EXPERIENCE

Research Intern (Applied ML, Smart Buildings)

Apr 2026 – Present

Columbus State University

Columbus, GA

- Continuing peer-reviewed energy-forecasting research 1-on-1 with Dr. Yesem Kurt Peker toward an AIRC follow-up publication on the ASHRAE GEP-III dataset (1,400 buildings, 2,377 building-meter series across electricity, chilled water, steam, and hot water).
- Surfaced a clean negative result through disciplined metric slicing (GMM soft-clustering lifted median per-meter R^2 but was net-harmful to aggregate R^2), then engineered per-meter state features (regime, Markov-transition, exponential-decay) that carry the gain with no clustering step at all.
- Current leader reaches overall test R^2 0.738 and median per-meter R^2 0.763 on a pruned 42-feature set, holding 0.738 to 0.725 across time splits where the clustering pipeline collapses to 0.002; that cross-split stability check is the headline claim of the paper.
- Own the full evaluation stack (paired Wilcoxon tests, bootstrap confidence intervals, controlled ablations, a five-check leakage audit, MLflow run tracking) on a local Python / Pandas / scikit-learn / XGBoost / tslearn workstation.

Data Science Research Assistant

Jun 2024 – Dec 2025

Columbus State University

Columbus, GA

- Led energy prediction for 91 buildings at Fort Moore (forecasted ~\$600K annual savings), lifting overall model R^2 from 0.82 to 0.98 (RMSE 12.16) via rolling Z-score outlier detection, STL imputation, and feature engineering, with XGBoost selected over Random Forest, LSTM, MSTL, and Prophet baselines.
- Authored the first-author IEEE Access 2026 paper end-to-end (DOI 10.1109/ACCESS.2026.3671169), formalising the outlier detection, STL imputation, XGBoost forecasting at $R^2 = 0.98$, and Bollinger Band fault detection developed in the role.
- Built end-to-end ETL/ELT pipelines on Azure Databricks integrating 4 heterogeneous sources into Azure SQL (medallion architecture, incremental loads, data-quality and freshness checks), consolidating 1.2M+ hourly time-series records, then drove a PySpark migration to scale from the 91-building pilot toward 300+ buildings.
- Proposed Bollinger Bands and implemented RMSE thresholding for energy fault detection; RMSE thresholding hit 97% accuracy with the lowest false-positive rate (0.021) in the co-authored AIRC 2025 paper benchmarking 4 methods on the LEAD dataset (1.48M readings).
- Student Dropout Prediction: XGBoost classifier on proprietary CSU registration data for coursework, highest accuracy (0.64) across 4 competing teams.

Data Engineer (Junior Software Engineer)

Innova Solutions

Jun 2022 – May 2023

Chennai, TN

- Shipped an automated PySpark/Scala schema-validation framework in 2 days for a problem unresolved for 6 weeks (3 sprints), enforcing field-level data contracts across 10+ production healthcare pipelines and removing a recurring class of downstream data-quality incidents.
- Built a modular Jackson (Scala) + Python converter that auto-generates Spark-compatible JSON schemas from any XML pipeline definition, handling deeply nested healthcare structures, repeating elements, optional vs. required fields, and namespace prefixes, cutting new-pipeline onboarding effort and defect rate.
- Flight Analysis: event-driven Spark pipeline on AWS where an S3 upload triggered Lambda/boto3 to spin up transient EMR clusters processing 130M+ commercial flight records for delay and cancellation metrics, torn down via Terraform to keep cost proportional to work.

PROJECTS

BirdCLEF+ 2026 | PyTorch, timm, torchaudio, PANNs CNN14, EfficientNet-B0, Mixup

kaggle.com/competitions/birdclef-2026 · github.com/kesiee/kaggle-birdclef-2026

- Kaggle multi-label audio classification with the Cornell Lab of Ornithology: identify bird, mammal, amphibian, and insect species in 5-second windows of field soundscapes under extreme class imbalance. Two-model ensemble of a CNN14 (PANNs, AudioSet-pretrained) frozen-embedding baseline and a headline EfficientNet-B0 fine-tuned on 128-mel log-spectrograms with Mixup (alpha 0.4), label smoothing, AdamW, and 5-fold stratified CV. Best fold reached macro ROC-AUC 0.9789 at epoch 27; CPU inference within the 9-hour Kaggle budget with time-shift TTA.

Store Sales Forecasting | Python, LightGBM, STL / MSTL, Prophet, Feature Engineering

kaggle.com/code/shashankkc/stl-forecasting · github.com/kesiee/kaggle-store-sales-forecasting

- Kaggle Store Sales Time Series challenge: daily unit-sales forecasts across thousands of (store, family) series for a large Ecuadorian grocery chain. Decomposition-first pipeline that scores seasonal strength per series (STL/MSTL) to gate model choice, engineers lag, rolling, oil-price, holiday, and calendar features on log1p(sales), and ensembles a per-family LightGBM, a global LightGBM, and an MSTL forecast. Best leaderboard RMSLE 0.56, improved from a 2.13 baseline through feature engineering and disciplined cross-validation.

March Machine Learning Mania 2026 | LightGBM, GMM, Elo Rating, Isotonic Calibration, Leave-One-Year-Out CV

github.com/kesiee/kaggle-march-madness-2026

- Kaggle NCAA tournament prediction (132,133 possible matchups, both genders) scored on Brier, so calibration matters as much as ranking. Elo-like dynamic ratings with between-season regression, a GMM over team-style features for soft cluster memberships, and a LightGBM matchup classifier on difference/absolute/sum feature forms with leave-one-year-out CV, blended 0.65 LGB / 0.35 Elo and passed through an isotonic calibrator. Validation Brier ~0.115 on the women's side with materially flatter mid-range calibration than v1.

gpt-activationNN: Transformer FFN activation study | PyTorch, Transformer Internals, Controlled A/B Design,

FLOPs Accounting, bf16 Mixed Precision

github.com/kesiee/gpt-activationNN

- Self-directed deep-learning research testing whether GELU in a Transformer's feed-forward block can be replaced with a learned function, then characterising what does win at matched compute. NanoGPT-style decoders (30M–58M params) trained on TinyStories under a controlled A/B protocol, every comparison compute-matched via explicit FLOPs and parameter accounting. Learned activations lost cleanly; a post-projection deep-gate sweep found gated_deep_silu at 5.673 val perplexity beating the size-matched GELU baseline (5.740). Negative results reported prominently with single-seed limitations flagged.

EDUCATION

Columbus State University

Aug 2023 – Dec 2025

M.S. in Applied Computer Science (Data Science)

GPA: 4.0/4.0

Bangalore Institute of Technology

Aug 2017 – Aug 2021

B.E. in Computer Science

F-1 STEM OPT authorized through 02/2029. No sponsorship required.

PUBLICATIONS

First-author: "An Integrated Data Engineering and Machine Learning Framework for Energy Prediction and Fault Detection in Smart Buildings," IEEE Access, 2026. doi:10.1109/ACCESS.2026.3671169

Co-author: "A Practical Framework for Energy Fault Detection in Smart Buildings," IEEE Xplore / AIRC 2025. doi:10.1109/AIRC64931.2025.11077504